1. Evaluate the following limits. Simplify your answers where possible.

a. $\lim_{x \to 27} \dfrac{x+27}{\sqrt[3]{x}+3}$

b. $\lim_{x \to 2} \dfrac{\frac{1}{x}-\frac{1}{2}}{x-2}$

c. $\lim_{x \to 0} \dfrac{(2+x)^3-8}{x}$

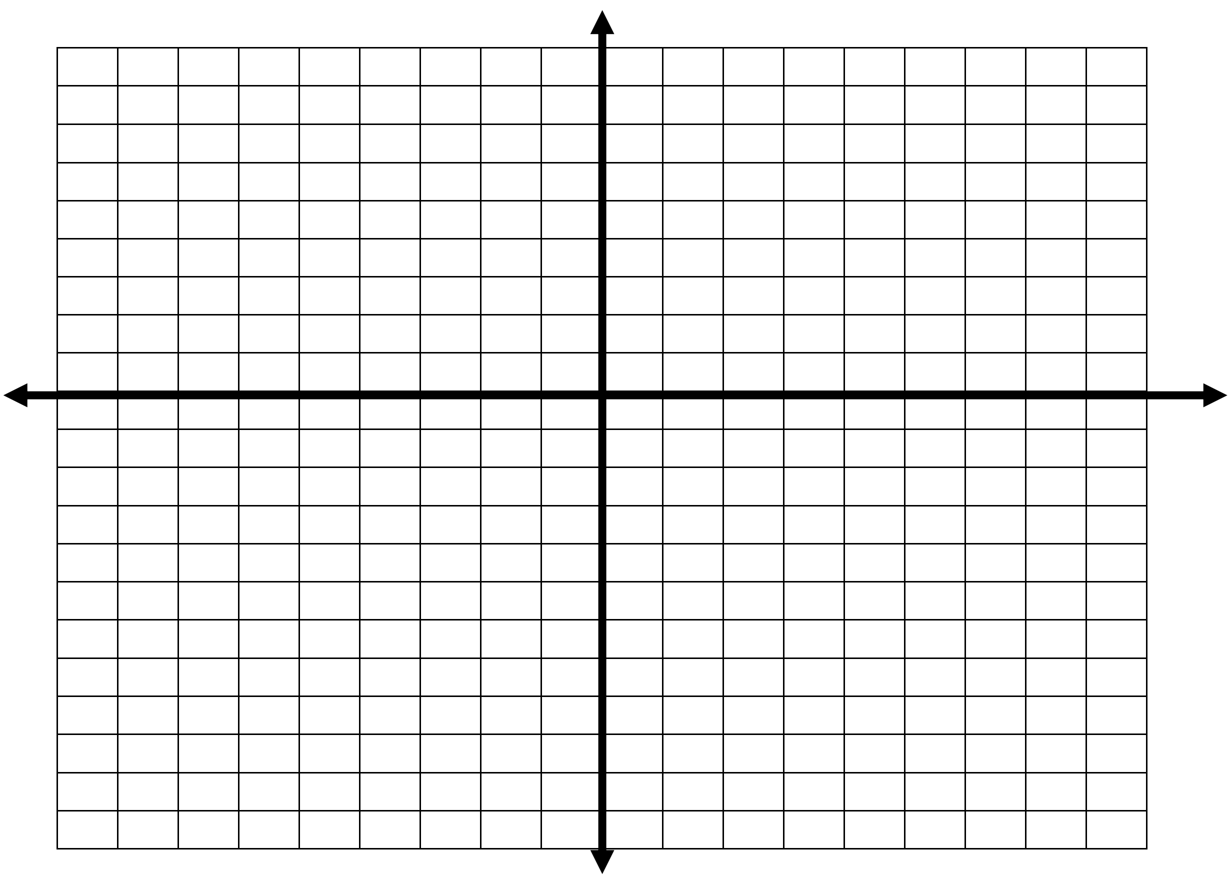
d. $\lim_{t \to 0} \dfrac{t}{\sqrt{1+3t}-1}$

e. $\lim_{t \to 0} \left(\dfrac{1}{t\sqrt{t+1}} - \dfrac{1}{t} \right)$

f. $\lim_{x \to 1} \dfrac{\sqrt[9]{x}-1}{\sqrt[3]{x}-1}$

2. Sketch a properly labelled graph of the following piecewise function.

$$f(x) = \begin{cases} 2-x^2 & , x \in (-\infty, 0) \\ x+3 & , x \in [0,1) \\ (-x+1)^2 & , x \in [1, \infty) \end{cases}$$



3. Determine whether or not the following function is continuous on the interval $(-\infty,\ \infty)$.
If the function is not continuous, state all value(s) of x where the function is discontinuous.

$$f(x)=\begin{cases} x^3-1 & ,x\in(-\infty,0) \\ -(4x+1) & ,x\in[0,1) \\ 2\sqrt{x}-7 & ,x\in[1,\infty) \end{cases}$$

4. Determine $\displaystyle\lim_{x\to 3}\frac{\sqrt{7-x}-2}{1-\sqrt{4-x}}$.

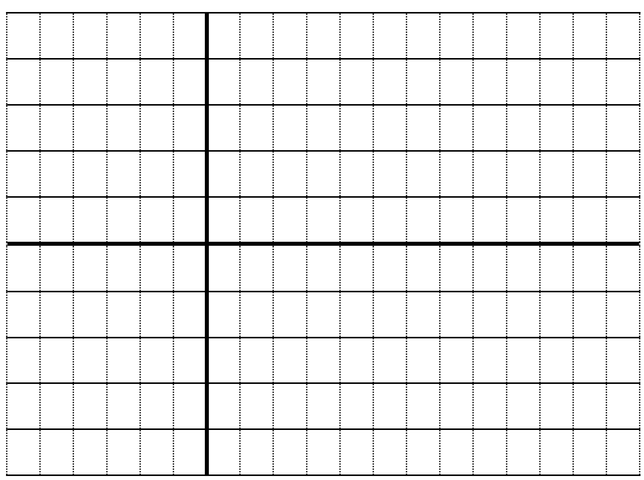
5. Determine all values of b and c in the function below, where $b, c\in\mathbb{R}$ such that $f(x)$ is continuous on the interval $(-\infty,\ \infty)$. Show all steps.

$$f(x)=\begin{cases} cx-2 & ,x\in(-\infty,-1) \\ -bx^2+2c & ,x\in[-1,1) \\ c\sqrt{x}+b+1 & ,x\in[1,\infty) \end{cases}$$

6. Sketch the graph of **any** function that satisfies the following conditions:

$f(0)=1$, $\displaystyle\lim_{x\to -3}f(x)=-2$, $\displaystyle\lim_{x\to 1^-}f(x)=0$, $\displaystyle\lim_{x\to 1^+}f(x)=2$, $\displaystyle\lim_{x\to\infty}f(x)=-1$

The function is continuous at all points in the interval $(-\infty,\ 1)\cup(1,\ \infty)$.



7. If $\displaystyle\lim_{x\to -1}\frac{x^3-ax^2-x+4}{x+1}$ exists, determine the value of a.

8. Determine **all** values of p and q such that $\displaystyle\lim_{x\to 0}\frac{\sqrt{px+q}-3}{x}=1$, where $p\ \&\ q\in\mathbb{R}$.